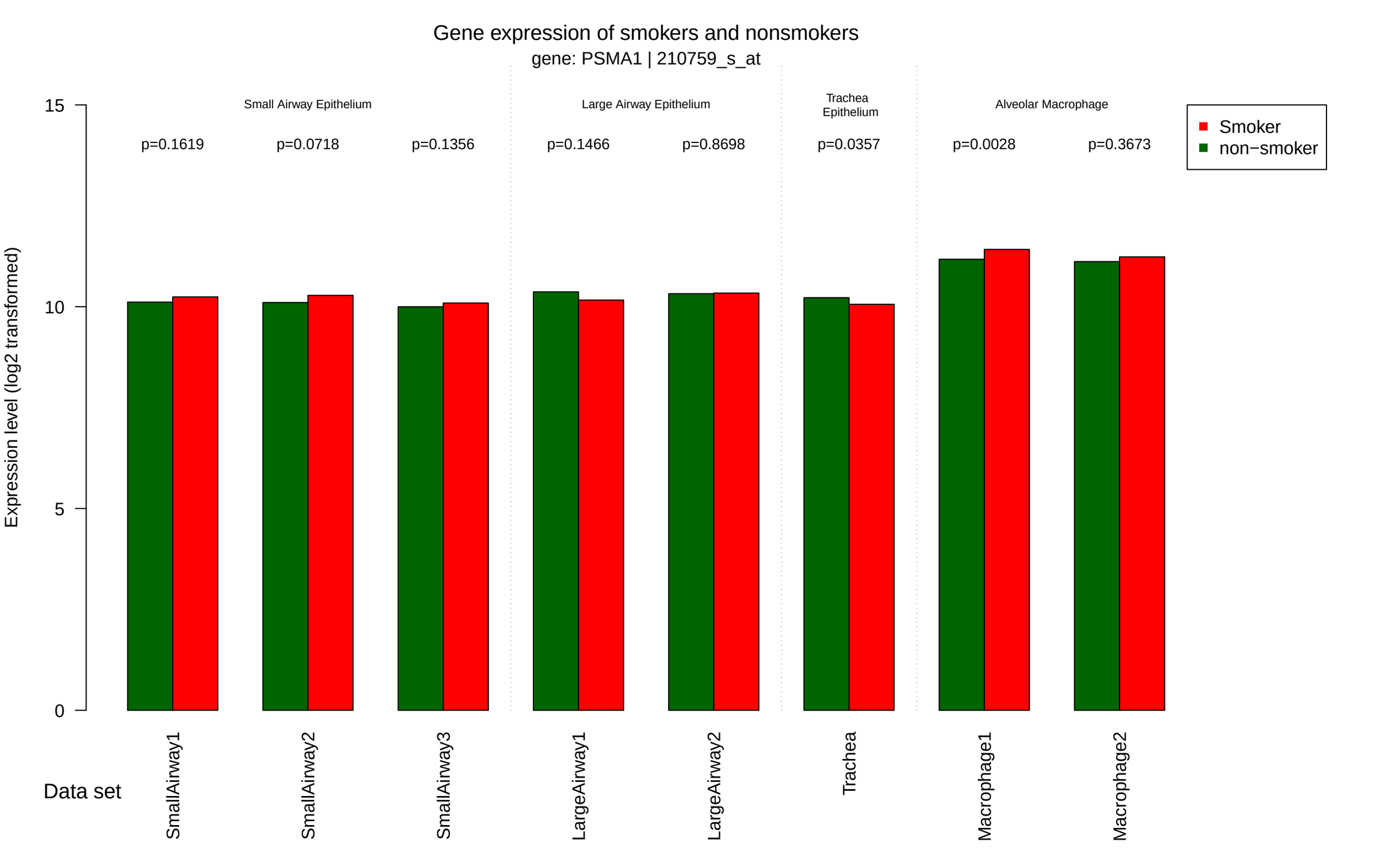
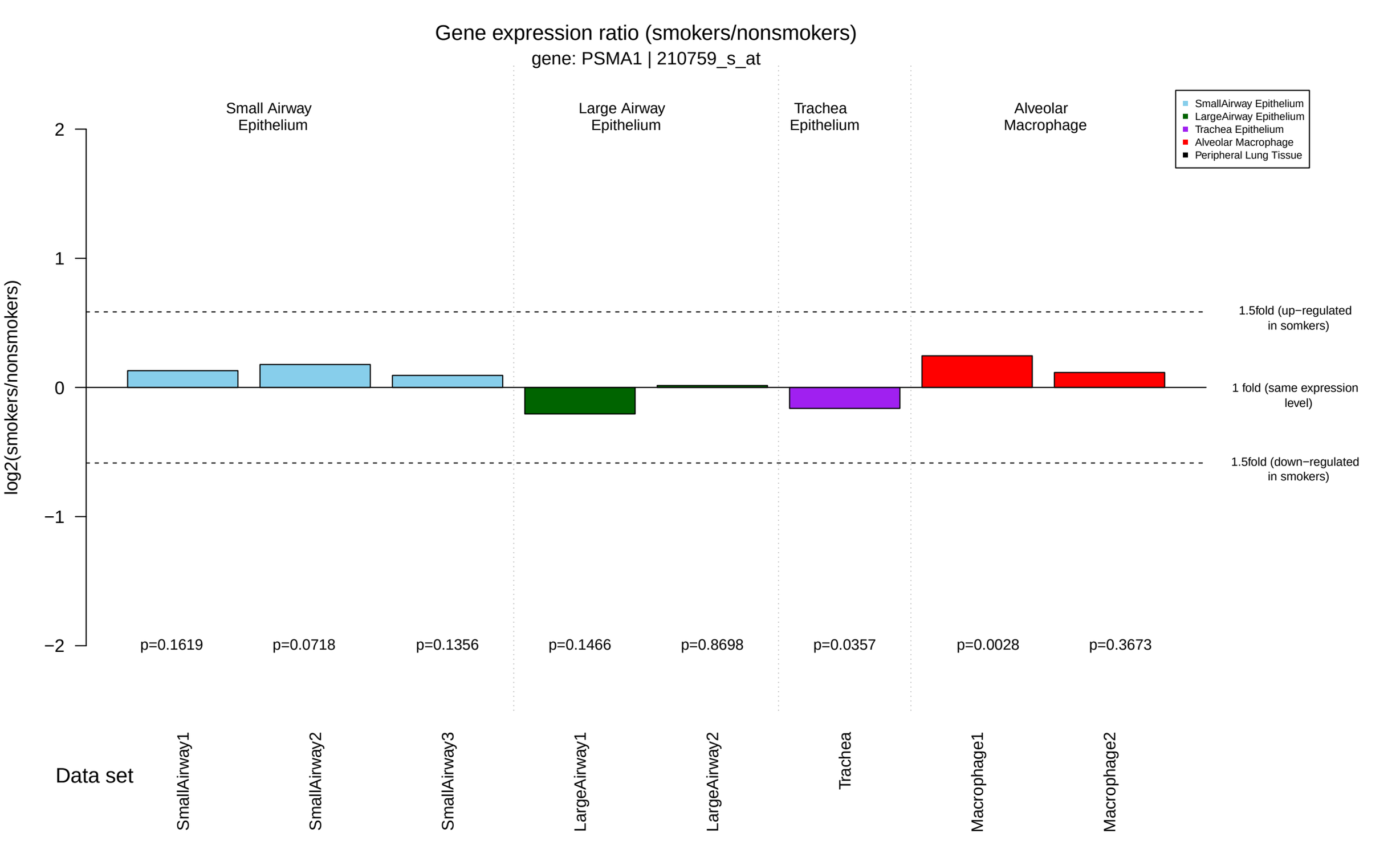
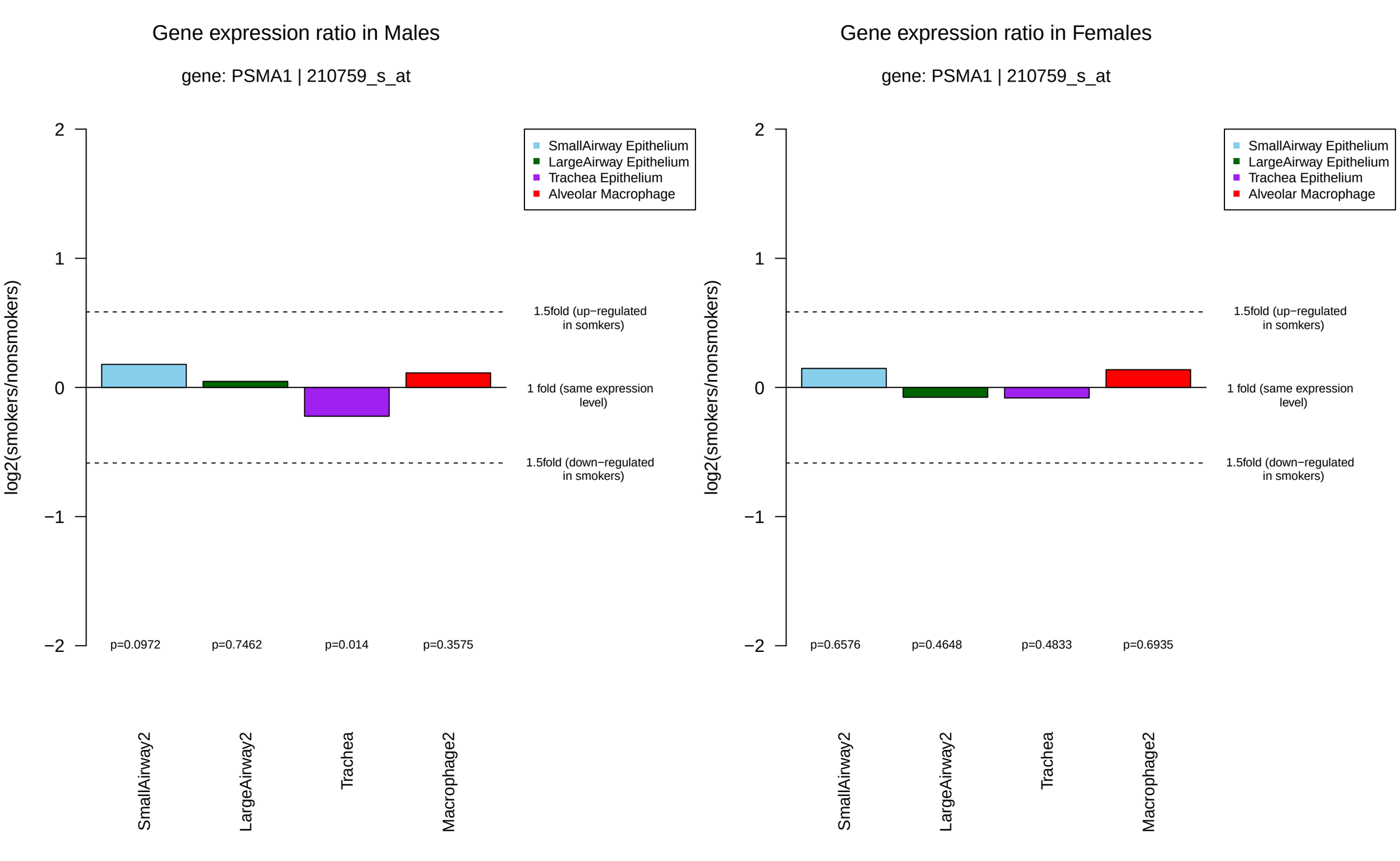
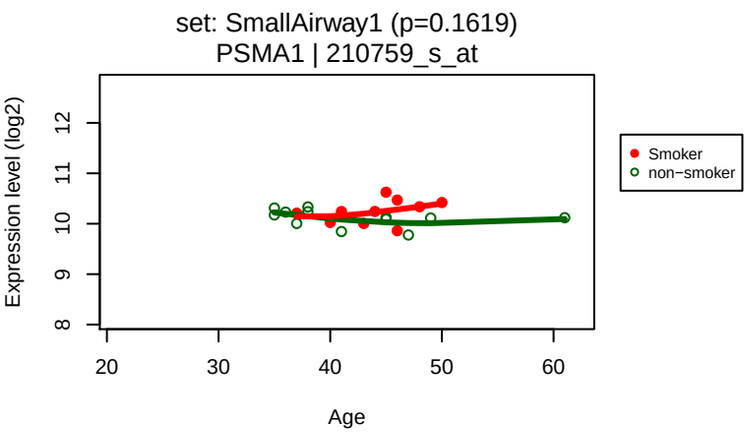
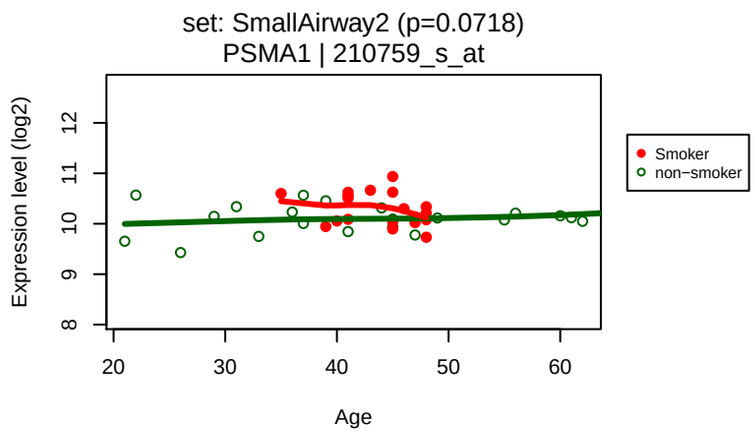
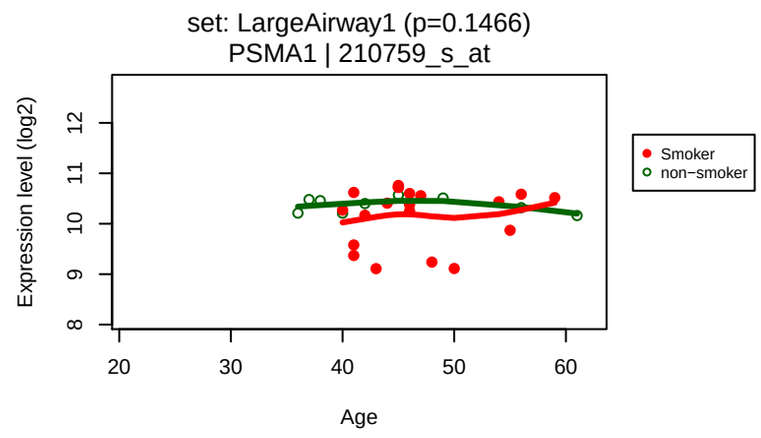

Gene expression of smokers and nonsmokers
gene: PSMA1 | 210759_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1619
p=0.0718
p=0.1356
p=0.1466
p=0.8698
p=0.0357
p=0.0028
p=0.3673
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

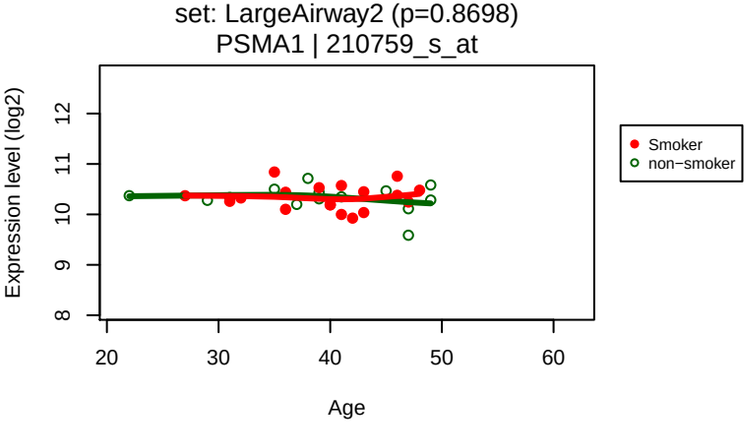

Gene expression ratio (smokers/nonsmokers)
gene: PSMA1 | 210759_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1619
p=0.0718
p=0.1356
p=0.1466
p=0.8698
p=0.0357
p=0.0028
p=0.3673
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

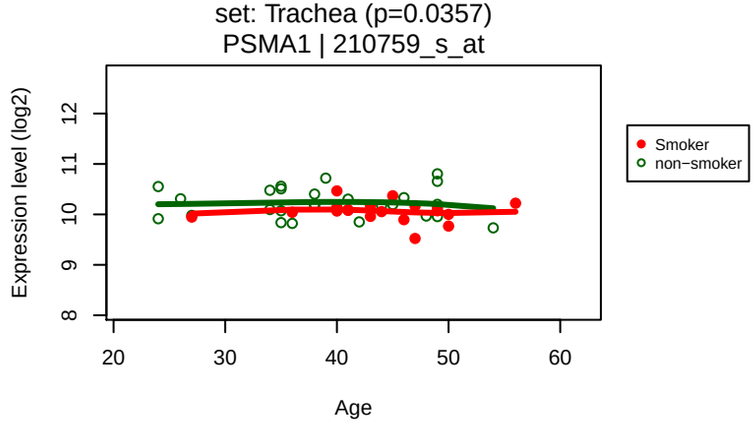

Gene expression ratio in Males
gene: PSMA1 | 210759_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0972
p=0.7462
p=0.014
p=0.3575
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA1 | 210759_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6576
p=0.4648
p=0.4833
p=0.6935
SmallAirway2
LargeAirway2
Trachea
Macrophage2

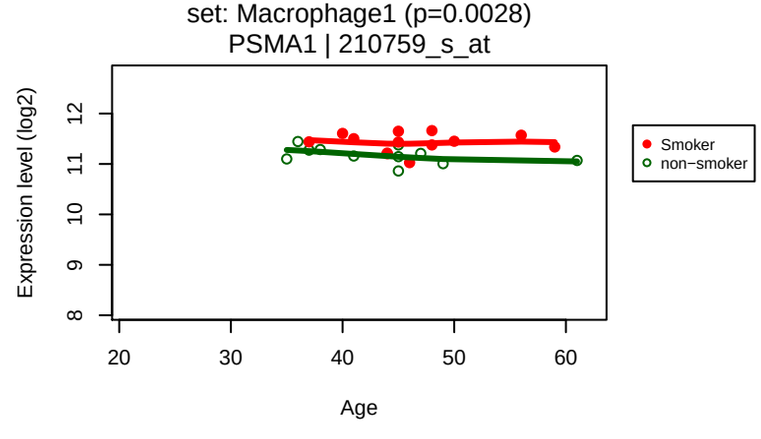

set: SmallAirway1 (p=0.1619)
PSMA1 | 210759_s_at
Expression level (log2)
Age
Smoker
non−smoker

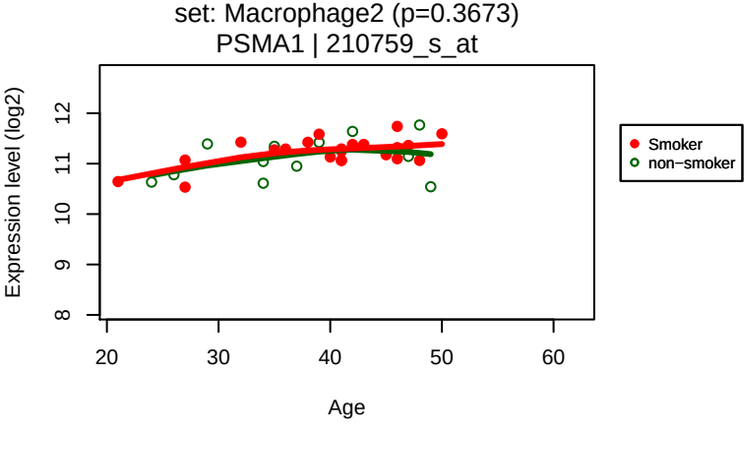

set: SmallAirway2 (p=0.0718)
PSMA1 | 210759_s_at
Expression level (log2)
Age
Smoker
non−smoker

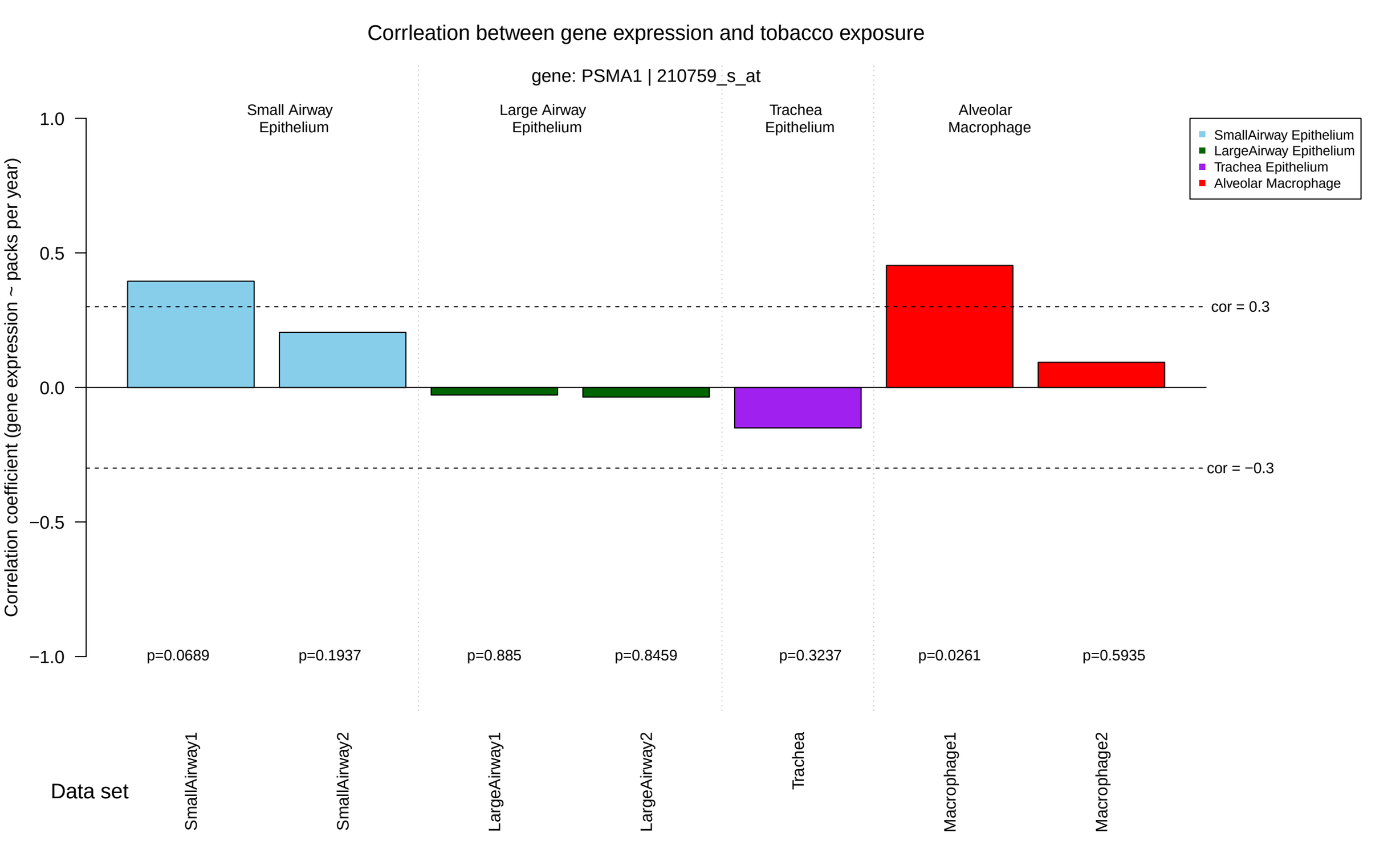

set: LargeAirway1 (p=0.1466)
PSMA1 | 210759_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8698)
PSMA1 | 210759_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0357)
PSMA1 | 210759_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0028)
PSMA1 | 210759_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3673)
PSMA1 | 210759_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA1 | 210759_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0689
p=0.1937
p=0.885
p=0.8459
p=0.3237
p=0.0261
p=0.5935
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2